Linux Drive Forensics: Complete Lock & Unlock Manual

Reference Guide for lsblk, blockdev, hdparm, NVMe, and Mount Operations

1. DRIVE IDENTIFICATION & STATUS INSPECTION (LSBLK)

`lsblk` lists block devices and displays whether a device is currently write-protected (Read-Only).

📋 List All Block Devices & Partitions

```
lsblk
```

Shows connected drives, sizes, names, and mount points.

🔍 Inspect Read-Only (RO) Column Status

```
lsblk -o NAME,SIZE,TYPE,RO,MOUNTPOINT
```

Displays the `RO` column: `1` = Write-Blocked (Read-Only), `0` = Unlocked (Read-Write).

> **Example Output:** `sdb 14.9G disk 1 /mnt/forensic` (RO = 1 indicates drive is locked).

🎯 Specific Disk RO Status Check

```
lsblk -d -o NAME,RO /dev/sdX
```

Inspects status for a single target disk without partition child rows.

> **Example:** `lsblk -d -o NAME,RO /dev/sdb`

2. KERNEL BLOCK DEVICE CONTROL (BLOCKDEV)

`blockdev` forces Linux kernel read-only or read-write flags on block devices.

🔒 Lock Drive (Enable Kernel Write-Blocking)

```
sudo blockdev --setro /dev/sdX
```

Sets kernel flag to Read-Only mode for target drive.

> **Example:** `sudo blockdev --setro /dev/sdb`

🔍 Query Kernel Lock Status

```
sudo blockdev --getro /dev/sdX
```

Returns `1` if locked (Read-Only) or `0` if unlocked (Read-Write).

> **Example:** `sudo blockdev --getro /dev/sdb` → Output: `1`

🔓 Unlock Drive (Disable Write-Blocking / Set Read-Write)

```
sudo blockdev --setrw /dev/sdX
```

Removes kernel read-only restriction, restoring read-write capabilities.

> **Example:** `sudo blockdev --setrw /dev/sdb`

3. SATA/ATA HARDWARE FLAGS (HDPARM)

`hdparm` manages driver and device flags directly for ATA/SATA drives.

🔒 Lock Drive (Set Read-Only Flag / Write-Block)

```
sudo hdparm -r1 /dev/sdX
```

Sets hardware driver read-only flag to ON.

> **Example:** `sudo hdparm -r1 /dev/sdc`

🔍 Check hdparm Read-Only Status

```
sudo hdparm -r /dev/sdX
```

Displays flag state. `readonly = 1 (on)` = Locked, `readonly = 0 (off)` = Unlocked.

> **Example:** `sudo hdparm -r /dev/sdc` → Output: `readonly = 1 (on)`

🔓 Unblock Drive (Set Read-Write / Reset Flag)

```
sudo hdparm -r0 /dev/sdX
```

Resets driver flag to 0, allowing write operations again.

> **Example:** `sudo hdparm -r0 /dev/sdc`

4. NVME SSDS & READ-ONLY MOUNT OPERATIONS

⚡ NVMe SSD Write-Block & Unlock

```
sudo blockdev --setro /dev/nvme0n1
```

For NVMe drives, use blockdev with namespace identifier. Unlock using `--setrw`.

📁 Read-Only Mount (Forensic Safe Mount)

```
sudo mount -o ro,noload /dev/sdX1 /mnt/target
```

Mounts partition in read-only mode without replaying journal log files (safe for EXT4/NTFS).

🔄 Remount Partition to Read-Write

```
sudo mount -o remount,rw /mnt/target
```

Remounts an active read-only mount point back to read-write mode.

5. QUICK COMMAND SUMMARY MATRIX

Tool	Lock Command (Write-Block)	Status Check Command	Unlock Command (Read-Write)
lsblk	N/A (Inspection Tool)	`lsblk -o NAME,RO`	N/A (Inspection Tool)
blockdev	`sudo blockdev --setro /dev/sdb`	`sudo blockdev --getro /dev/sdb`	`sudo blockdev --setrw /dev/sdb`
hdparm	`sudo hdparm -r1 /dev/sdb`	`sudo hdparm -r /dev/sdb`	`sudo hdparm -r0 /dev/sdb`
mount	`sudo mount -o ro,noload ...`	`mount | grep /mnt/target`	`sudo mount -o remount,rw ...`